

Activity: APP BUILDING USING WEKA TOOL

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1. Weather-Based Play Prediction

Build a WEKA application that predicts whether a person should play outdoors based on weather conditions.

- Dataset: Outlook, Temperature, Humidity, Windy
- Target: Play (Yes/No)
- Algorithm: **J48 Decision Tree**
- Tasks:
 1. Load the dataset into WEKA.
 2. Preprocess the data.
 3. Apply J48.
 4. Visualize the decision tree.
 5. Predict Play for a new weather condition.

2. Flower Species Classification 🌸

Build a WEKA application that identifies the species of a flower from its physical measurements.

- Dataset: Iris dataset
- Attributes: Sepal Length, Sepal Width, Petal Length, Petal Width
- Target: Iris Species
- Algorithm: **J48 or Naive Bayes**
- Tasks:
 1. Load the Iris dataset.
 2. Select the classification algorithm.
 3. Train the model.
 4. Evaluate its accuracy.

5. Classify a new flower.

Answers:

1) **Weather-Based Play Prediction**

Classifier output

Model attributes

Incoming attributes

```

(numeric) Weight    --> 1 (numeric) Weight
(nominal) Color      --> 2 (nominal) Color
(nominal) Size       --> 3 (nominal) Size
(numeric) Firmness   --> 4 (numeric) Firmness
(nominal) Quality     --> 5 (nominal) Quality

```

Time taken to build model: 0 seconds

=== Predictions on test set ===

```

inst#,actual,predicted,error,prediction
1,1:?,1:Good,,1
2,1:?,2:Bad,,1

```

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.01 seconds

=== Summary ===

Total Number of Instances

0

Ignored Class Unknown Instances

2

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	?	?	?	?	?	?	?	?	Good
	?	?	?	?	?	?	?	?	Bad
Weighted Avg.	?	?	?	?	?	?	?	?	

=== Confusion Matrix ===

```

a b  <-- classified as
0 0 | a = Good
0 0 | b = Bad

```

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose J48 - C 0.25 - M 2

Test options

☐ Use training set
 ☒ Supplied test set

Set...

☐ Cross-validation

Folds 10

☐ Percentage split

% 66

More options...

(Nom) Quality

Start

Stop

Result list (right-click for options)

11:25:52 - trees.J48

11:45:17 - misc.InputMappedClassifier

11:50:21 - misc.InputMappedClassifier

Classifier output

J48 pruned tree

```

Color = Red: Good
Color = Green: Bad
Color = Yellow: Good

```

Number of Leaves

Size of the tree

Time taken to build

=== Stratified cross

=== Summary ===

Correctly Classified

Incorrectly Classified

Kappa statistic

Mean absolute error

Root mean squared error

Relative absolute error

Root relative squared error

Total Number of Instances

=== Detailed Accuracy By Class ===

Weighted Avg.

=== Confusion Matrix ===

```

a b  <-- classified as
11 0 | a = Good
1 8 | b = Bad

```

Weka Classifier Tree Visualizer: 11:25:52 - trees.J48 (fruit_quality)

Tree View

```

graph TD
    Color((Color)) -- "- Red" --> Good50[Good (5 0)]
    Color -- "- Green" --> Bad00[Bad (0 0)]
    Color -- "- Yellow" --> Good00[Good (0 0)]

```

2) Flower Species Classification

Classifier output

=== Run information ===

Scheme: weka.classifiers.trees.J48 -C 0.25 -M 2
 Relation: iris
 Instances: 150
 Attributes: 5
 SepalLength
 SepalWidth
 PetalLength
 PetalWidth
 Class
 Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree

```
PetalWidth <= 0.6: Iris-setosa (50.0)
PetalWidth > 0.6
|   PetalWidth <= 1.7
|   |   PetalLength <= 4.9: Iris-versicolor (48.0/1.0)
|   |   PetalLength > 4.9
|   |   |   PetalWidth <= 1.5: Iris-virginica (3.0)
|   |   |   PetalWidth > 1.5: Iris-versicolor (3.0/1.0)
|   PetalWidth > 1.7: Iris-virginica (46.0/1.0)
```

Number of Leaves : 5

Size of the tree : 9

Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	144	96	%
Incorrectly Classified Instances	6	4	%
Kappa statistic	0.94		

Mean absolute error	0.035
Root mean squared error	0.1586
Relative absolute error	7.8705 %
Root relative squared error	33.6353 %
Total Number of Instances	150

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.980	0.000	1.000	0.980	0.990	0.985	0.990	0.987	Iris-setosa
	0.940	0.030	0.940	0.940	0.940	0.910	0.952	0.880	Iris-versicolor
	0.960	0.030	0.941	0.960	0.950	0.925	0.961	0.905	Iris-virginica
Weighted Avg.	0.960	0.020	0.960	0.960	0.960	0.940	0.968	0.924	

=== Confusion Matrix ===

```
a b c  <-- classified as
49 1 0 | a = Iris-setosa
0 47 3 | b = Iris-versicolor
0 2 48 | c = Iris-virginica
```

